

4.1

$$a. SST = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n \bar{y}^2$$

$$\Rightarrow SST = 7825.94 - 20 \cdot (119.21)^2 \\ = 445.458$$

$$R^2 = \frac{375.47}{445.458} \approx 0.843$$

b.

$$R^2 = 1 - \frac{SSE}{SST}$$

$$\Rightarrow SSE = (1 - R^2) SST \approx 151.65$$

$$\hat{\sigma}^2 = \frac{SSE}{n-2}$$

$$\Rightarrow \hat{\sigma}^2 = \frac{151.65}{18} \approx 8.425$$

c.

$$\sum \hat{e}_i^2 = SSE$$

$$\Rightarrow R^2 = 1 - \frac{182.55}{631.61} \approx 0.7105$$

4.3

a. The fitted least squares line: $\hat{y}_i = 1.2 + 0.8 x_i$

$$x_0 = 4 \Rightarrow \hat{y}_0 = 4.4$$

$$b. se(f) = s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$\bar{x} = 1$$

$$s = \sqrt{1.2}$$

$$\Rightarrow se(f) = \sqrt{1.2} \cdot \sqrt{\frac{1}{5} + \frac{(4-1)^2}{10}} \approx 1.1489$$

$$c. \quad S_{\text{pred}} = s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$\Rightarrow S_{\text{pred}} = \sqrt{1.2} \cdot \sqrt{1 + \frac{1}{5} + \frac{(4-1)^2}{10}} \approx 1.5875$$

95% P.I. for $\hat{y}_0$

$$\hat{y}_0 \pm t_{0.025, 5-2} \cdot S_{\text{pred}}$$

$$= [-1.052, 9.052]$$

d. 99% P.I. for $\hat{y}_0$:

$$\hat{y}_0 \pm t_{0.005, 5-2} \cdot S_{\text{pred}}$$

$$= [-5.272, 13.272]$$

e. $\bar{x} = 1$

$$\Rightarrow S_{\text{pred}} = \sqrt{1.2} \cdot \sqrt{1 + \frac{1}{5} + \frac{(1-1)^2}{10}} = 1.2$$

95% P.I. for y given $x = \bar{x}$:

$$\bar{y} \pm t_{0.025, 5-2} \cdot S_{\text{pred}}$$

$$= [-2.819, 4.819]$$